Syllabus Year: 2023-2024

CHRIST (Deemed to be University), Bengaluru - 560029

End Semester Examinations - November 2025

Bachelor of Computer Applications SEMESTER - II

Course Code: BCA104-2 **Max Marks: 100**

Course Name: DISCRETE MATHEMATICS **Duration: 03:00**

General Instructions:

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / course title and number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer book. No sheets should be detached from the answer book.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer book.

Course Outcome: Students will be able to:

CO1: Demonstrate a working knowledge of set notation and elementary set theory and recognize the connection between set operations and logic.

CO2: Apply the different properties of injections, surjections, bijections, compositions, and inverse functions.

CO3: Understand different types of matrices and operations among matrices.

CO4: Determine when a relation is reflexive, symmetric, antisymmetric, transitive, apply the properties of equivalence relations and partial orderings, and explain the connection between equivalence relations.

CO5: Effective express the elementary concepts and results of Number theory.

Section A **3*8=24**

1)	Define a set and provide an example of each type: finite, infinite, empty, singleton. [CO1]	**3**	(CO1, L2)
2)	What is the power set of {a, b}? [CO1]	**3**	(CO1, L2)
3) a)	Given sets A = {1, 2, 3} and B = {4, 5}, find the Cartesian product A × B. [CO1]	**3**	(CO1, L2)
	OR		
b)	If C = {x, y} and D = {z}, find the Cartesian product C × D. [CO1]	**3**	(CO1, L2)
4)	Prove that if f: A → B is a function and A is finite, then the range of f is also finite. [CO1]	**3**	(CO1, L3)
5)	Discuss the properties of a relation R on set A = {1, 2, 3} defined by R = {(x, y) | x + y is even}. Is R reflexive? Symmetric? Transitive? [CO4]	**3**	(CO4, L3)
6)	Define a matrix and provide an example of a 2x3 matrix. [CO3]	**3**	(CO3, L1)
7)	Given the matrices A = egin{pmatrix} 1 & 2 \ 3 & 4 \ dotdotdot end{pmatrix} and B = egin{pmatrix} -1 & 0 \ 0 & 1 \ dotdotdot end{pmatrix}, find the product AB. [CO3]	**3**	(CO3, L2)
8)	Prove that if f: A → B is a surjective function and |A| = |B|, then f must be bijective. [CO2]	**3**	(CO2, L4)

Section B

8*7=56

9)	Given the sets A = {1, 2, 3} and B = {a, b}, find all possible functions from A to B. [CO1]	**8**	(CO1, L2)
10)	Define an equivalence relation and provide an example. Prove that the relation is reflexive, symmetric, and transitive. [CO4]	**8**	(CO4, L3)
11)	Given the matrix A = egin{pmatrix} 2 & -1 \ 4 & -2 \ dotdotdot end{pmatrix}, find its inverse if it exists. [CO3]	**8**	(CO3, L4)
12)	Define a prime number and prove that there are infinitely many prime numbers. [CO5]	**8**	(CO5, L4)
13)	Prove that the function f: Z → Z defined by f(x) = x^2 is neither injective nor surjective. [CO2]	**8**	(CO2, L4)
14)	Given the relation R on set A = {1, 2, 3} defined by R = {(x, y) | x + y is even}, determine if R is an equivalence relation. [CO4]	**8**	(CO4, L3)

Section C

10*2=20

15)	Prove that for any integer a, if a^2 is even then a is also even. [CO5]	**10**	(CO5, L4)
16)	Given the sets A = {1, 2, 3} and B = {a, b}, find all possible relations from A to B. Prove that a relation is reflexive if for every x in A, (x, x) is in the relation. [CO4]	**10**	(CO4, L3)
17)	Define a matrix operation and provide an example. Prove that the transpose of a matrix A, denoted by A^T, satisfies (A^T)^T = A. [CO3]	**10**	(CO3, L4)

Revised Bloom's Taxonomy (RBT) Levels

L1 - Remember
L2 - Understand
L3 - Apply
L4 - Analyse
L5 - Evaluate
L6 - Create

AI-assisted draft - teacher review and approval required before use